

Hotel Bookings Analysis Report

Dataset: hotel_bookings_final.csv

Key Observations

Q1. Key Trends and Patterns

Trend 1: Room Type and Cancellation Behavior

- **Standard rooms have the highest cancellation rate (23.3%)**, while Deluxe (16%) and Suite (17.98%) rooms are cancelled less frequently.
- Interpretation: Customers booking lower-cost rooms are more price-sensitive and less committed, making them more likely to cancel. Premium rooms show higher commitment and stability.

Trend 2: Star Rating Influence

- **Cancellation rates are slightly higher for 5-star properties (21.3%)** compared to 2–4-star properties (~20%).
- Interpretation: High-value bookings carry slightly higher risk, potentially due to flexible travel plans or high-cost investments, whereas mid-range bookings are more predictable.

Trend 3: Seasonal/Temporal Patterns

- **Booking volumes peak in April (4494 bookings)**, while low-demand months include February, June, and September (~2200 bookings).
- **Highest cancellation rates occur in July (30.33%), August (28.77%), and December (25.76%), while April (16.31%) and November (15.93%) see the lowest.**
- Interpretation: Peak travel months may see more cancellations due to last-minute plan changes. Low-demand months show stable booking behavior.

Additional Trends

- Booking Channels: **Cancellation rates are similar** across Web (20.33%), iOS (20.37%), and Android (20%), showing that channel itself is not a major driver.
- Coupon Usage: **Bookings made with coupons have slightly higher cancellations (20.8%)** than those without (20.1%), indicating discount-driven customers are marginally more likely to cancel.
- Lead Time & Stay Type: Average lead time (~30 days) and stay type (Business vs Leisure) show minimal impact on cancellations.

Q2. Booking Channels

- Majority of bookings are made via **Web**, followed by **iOS** and **Android**.
- Revenue distribution is roughly proportional to booking volume.
- **Cancellation rates are similar across all channels (~20%)**, indicating channel itself is **not a major driver** of cancellations.

Room Type

- **Standard rooms** have the **highest cancellation rate (23.3%)**.
- **Deluxe (16%)** and **Suite (17.98%)** bookings are less likely to be cancelled.
- Lower-cost rooms are more sensitive to changes in travel plans, indicating **price-sensitive behavior**.

Star Rating

- Cancellation rates are slightly higher for **5-star properties (21.3%)**, compared to 2–4-star properties (~20%).
- Premium guests tend to cancel slightly more, possibly due to **flexible travel plans or high-value bookings**.

Lead Time

- Average lead time is ~30 days for both cancelled and confirmed bookings.
- Lead time alone is **not a significant factor** in cancellations.

Stay Type

- Cancellation rates are similar for **Business (20.05%)** and **Leisure (20.35%)** stays.
- Both segments exhibit comparable cancellation behavior, so interventions can be applied broadly.

Coupon Usage

- Bookings made with coupons have slightly higher cancellation rates (**20.8%**) compared to those without (**20.1%**).
- Discount-driven customers are marginally more likely to cancel.

Seasonal / Temporal Trends

- Peak booking month: **April (4494 bookings)**.
- Low-demand months: **February, June, September (~2200 bookings)**.

- Highest cancellation months: **July (30.33%), August (28.77%), December (25.76%).**
- Lowest cancellation months: **April (16.31%), November (15.93%).**
- Seasonal trends indicate that **peak demand periods may see higher cancellations due to plan changes**, while off-peak months have stable behavior.

Q3. What Do We Notice About Cancellation Behavior?

1. **Room Type is a Key Driver:** Standard rooms and discounted bookings are more likely to be cancelled, while premium rooms are more committed.
2. **Seasonality Matters: Cancellations spike during peak travel months (July, August, December), likely due to plan changes or high booking volumes.**
3. **Channel, Lead Time, and Stay Type Are Minor Factors:** Cancellation rates are consistent across channels, lead times, and stay types, suggesting that these factors do not significantly influence behavior.
4. **Discounts Slightly Increase Risk: Coupon-driven bookings are slightly more prone to cancellations**, highlighting the need for strategic discount policies.

Cancellation behavior is influenced more by room type, star rating, seasonality, and discounts than by booking channel, lead time, or stay type. Businesses should focus interventions on high-risk segments to reduce cancellations and improve profitability.

Root Cause Analysis

Q1. Cancellation Patterns

1. **Room Type / Price Sensitivity:** Standard rooms are cancelled most frequently (23.3%), indicating **budget-conscious customers are more likely to cancel.**
2. **Premium Bookings:** 5-star properties have slightly higher cancellations (21.3%) — high-value bookings may be more flexible or prone to last-minute changes.
3. **Coupon Usage:** Slightly higher cancellations for coupon users (20.8%) suggest **discount-driven bookings carry slightly higher risk.**
4. **Lead Time & Stay Type:** Both lead time (~30 days) and stay type show negligible influence on cancellations.

Q2. Performance by Property Type and Channel

i. Property Type Performance

- **Premium rooms (Deluxe and Suite)** consistently show **lower cancellation rates** (16–18%) compared to Standard rooms (23.3%).
- **Reasons for better performance:**
 1. **Higher financial commitment:** Guests paying more for premium rooms are less likely to cancel because of the higher cost.
 2. **Purpose-driven bookings:** Deluxe and Suite bookings are often linked to **special occasions, business trips, or vacations**, making them less flexible.
 3. **Perceived value & satisfaction:** Premium rooms usually offer better facilities, service quality, and customer experience, which increases **booking adherence**.

Implication: Promoting premium rooms can **stabilize revenue** and reduce cancellation risk, while Standard rooms require additional interventions.

ii. Booking Channel Performance

- Analysis shows that **Web, iOS, and Android channels have nearly identical cancellation rates (~20%)**.
- **Interpretation:** Differences in performance are mainly due to **booking volume**, not cancellation behavior.
 - Web may have more total bookings due to accessibility, but per-booking cancellation risk is similar to mobile apps.
 - This indicates that **channel-specific policies are less critical**, and focus should instead be on high-volume channels for marketing and promotions.

Q3. Seasonal / Temporal Trends

- **Peak months (July, August, December)** exhibit the **highest cancellation rates** (28–30%), likely due to:
 1. High booking volumes during holidays and peak travel seasons.
 2. Guests making early reservations and later altering plans.
- **Off-peak months (April, November)** show the **lowest cancellation rates** (~16%), indicating more committed travelers or stable demand.
- **Business Implications:**
 1. Use **dynamic pricing and high-value promotions** during peak months to maximize revenue.

2. Implement **flexible booking policies and incentives** during off-peak months to encourage stable bookings.
 3. Seasonal analysis aids in **resource allocation, staffing, and inventory management**, ensuring operational efficiency.
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Business Recommendations

Q1. Reducing Cancellations

- **Focus on high-risk segments:** Standard rooms and coupon-driven bookings.
 - Implement **partial prepayment** or small non-refundable deposits.
 - Send **reminder emails, booking confirmations, or incentives** to encourage adherence.
- **Target premium bookings:** Introduce **early confirmation incentives or loyalty benefits** for 5-star bookings to reduce cancellations and increase repeat business.

Q2. Improving Profitability

- **Promote higher-star properties** to increase revenue per booking while lowering cancellation risk.
- **Upsell longer stays or premium rooms** to maximize revenue per customer.
- **Encourage repeat bookings** via loyalty programs, app-exclusive offers, and personalized campaigns.

Q3. Optimizing Pricing, Promotions, and Channel Strategy

- **Dynamic pricing:** Adjust room rates according to demand, season, and occupancy trends to maximize revenue.
- **Seasonal promotions:** Offer attractive deals during low-demand months to boost bookings while managing cancellation risk.
- **Coupon strategy:**
 - Offer **partial prepayment discounts** for coupon users to reduce risk.
 - Avoid heavy discounts for high-risk room types (Standard) to protect revenue.
- **Channel focus:** Since cancellation rates are similar across channels, concentrate **marketing and engagement on high-volume channels** rather than worrying about cancellation differences.

Conclusion

The analysis provides a **comprehensive view of booking patterns, cancellations, and revenue performance**:

- **High-risk areas:** Standard rooms, coupon bookings, and peak months with high cancellations.
- **Stable revenue sources:** Premium room types, off-peak months, and loyal customers.
- **Actionable strategies:** Partial prepayment, targeted promotions, dynamic pricing, loyalty programs, and seasonal planning.

By focusing on high-risk room types, seasonal trends, and discount bookings, the business can **reduce cancellations, optimize revenue, and improve profitability**, while maintaining a balanced, data-driven approach to marketing and operations.
